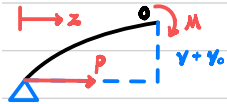


Theory

Imperfections



$$M(0), \quad M = P(v + v_0) = -EI \frac{d^2 v}{dz^2}$$

$$\therefore EI \frac{d^2 v}{dz^2} + Pv = -Pv_0 \quad \therefore \frac{d^2 v}{dz^2} + \mu^2 v = -\mu^2 v_0 = -\mu^2 v_{c0} \sin\left(\frac{\pi z}{L}\right) \quad \text{where } \mu = \frac{P}{EI}$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) + C \sin\left(\frac{\pi z}{L}\right) + D \cos\left(\frac{\pi z}{L}\right)$$

$$\therefore \frac{d^2 v}{dz^2} = -\mu^2 [A \sin(\mu z) + B \cos(\mu z)] - \left(\frac{\pi}{L}\right)^2 [C \sin\left(\frac{\pi z}{L}\right) + D \cos\left(\frac{\pi z}{L}\right)]$$

$$\text{substitute back, } \left(\mu^2 - \frac{\pi^2}{L^2}\right) [C \sin\left(\frac{\pi z}{L}\right) + D \cos\left(\frac{\pi z}{L}\right)] = -\mu^2 v_{c0} \sin\left(\frac{\pi z}{L}\right)$$

$$\therefore C = \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2} v_{c0}, \quad D = 0$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) + \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2} v_{c0} \sin\left(\frac{\pi z}{L}\right)$$

$$\text{At } z = 0, \quad v = 0$$

$$B = 0$$

$$\text{At } z = L, \quad v = 0$$

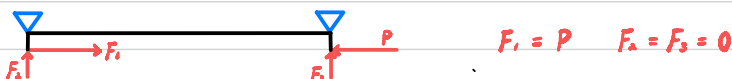
$$A \sin(\mu L) + \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2} v_{c0} \sin(\pi) = A \sin(\mu L) = 0$$

$$\therefore \sin(\mu L) = 0$$

$$\text{Let } \mu L = \pi \quad \therefore (\mu L)^2 = \frac{P}{EI} L^2 = \pi^2 \quad \therefore P_{crit} = \frac{\pi^2 EI}{L^2}$$

$$\text{At } z = \frac{L}{2}, \quad v_c = \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2} v_{c0} \sin\left(\frac{\pi}{2}\right) = \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2} v_{c0} = \frac{v_{c0}}{\frac{P_{crit}}{P} - 1} = P_{crit} \left(\frac{v_{crit}}{P}\right) - v_{c0}$$

Eccentricity



$$M(0), \quad M = P(v + e) = -EI \frac{d^2 v}{dz^2}$$

$$\therefore EI \frac{d^2 v}{dz^2} + Pv = -Pe \quad \therefore \frac{d^2 v}{dz^2} + \mu^2 v = -\mu^2 e \quad \text{where } \mu^2 = \frac{P}{EI}$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) + C$$

$$\therefore \frac{d^2 v}{dz^2} = -\mu^2 [A \sin(\mu z) + B \cos(\mu z)]$$

$$\text{substitute back, } \mu^2 C = -\mu^2 e \quad \therefore C = -e$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) - e$$

$$\text{At } z=0, v=0$$

$$\therefore 0 = B - e \quad \therefore B = e \quad \therefore A \sin(\mu z) + e [\cos(\mu z) - 1]$$

$$\text{At } z=L, v=0$$

$$\therefore 0 = A \sin(\mu L) + e [\cos(\mu L) - 1] \quad \therefore A = \frac{e[1 - \cos(\mu L)]}{\sin(\mu L)}$$

$$\therefore \sin(\mu L) = \cos(\mu L) - 1 = 0 \quad \therefore \mu L = n\pi$$

$$\text{Let } \mu L = \pi \quad \therefore (\mu L)^2 = \frac{P}{EI} L^2 = \pi^2 \quad \therefore P_{crit} = \frac{\pi^2 EI}{L^2}$$

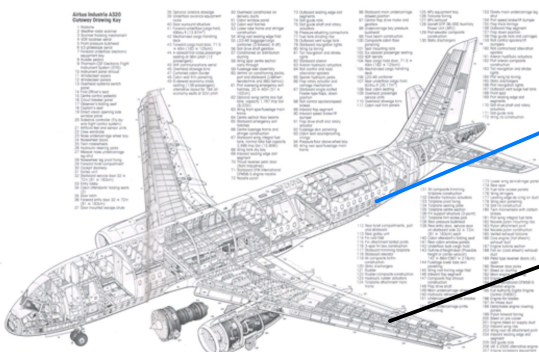
$$\text{At } z = \frac{L}{2}, v_c = A \sin\left(\frac{\mu L}{2}\right) + e [\cos\left(\frac{\mu L}{2}\right) - 1] = e \left(\frac{1 - \cos(\mu L)}{\sin(\mu L)} \sin\left(\frac{\mu L}{2}\right) + \cos\left(\frac{\mu L}{2}\right) - 1 \right)$$

$$= e \left(\frac{2 \sin^2\left(\frac{\mu L}{2}\right)}{2 \sin\left(\frac{\mu L}{2}\right) \cos\left(\frac{\mu L}{2}\right)} \sin\left(\frac{\mu L}{2}\right) + \cos\left(\frac{\mu L}{2}\right) - 1 \right) = e \left(\frac{1}{\cos\left(\frac{\mu L}{2}\right)} - 1 \right) = e \left[\frac{1}{\cos\left(\frac{\pi}{2} \sqrt{\frac{P}{P_{crit}}}\right)} - 1 \right]$$

Introduction

Objectives

- calculate P_{crit}
- investigate the effect of changing L, E, A, e and P_{trans}
- compare with theory

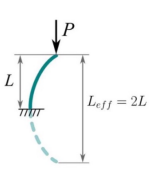


slender longerons

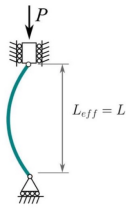
stringers

Theory

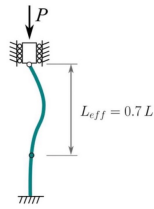
a) fixed-free end



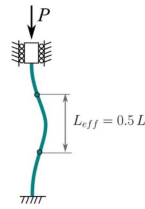
b) pin-pin end



c) fixed-pin end



d) fixed-fixed end



$$I = \frac{bd^3}{12}$$

$$P_{crit} = \frac{\pi^2 EI}{L_{eff}^2}$$

Assumptions

- struct is homogeneous E constant
- struct is prismatic A and I constant
- struct is perfectly straight no eccentricity
- compressive load applied to natural axis

11/27/2023 Euler Buckling During lab

Procedure

Safety

Parameters

Material	Young's Modulus (Nominal)
Steel	207 GN.m ⁻² (207 GPa)
Aluminium	69 GN.m ⁻² (69 GPa)
Brass	105 GN.m ⁻² (105 GPa)

from SM1005 User Guide

Procedure

- Strut 1 - Steel, 20 mm x 3 mm x 750 mm
- Strut 2 - Steel, 20 mm x 3 mm x 700 mm
- Strut 3 - Steel, 20 mm x 3 mm x 650 mm
- Strut 4 - Steel, 20 mm x 3 mm x 625 mm
- Strut 5 - Steel, 20 mm x 3 mm x 600 mm
- Strut 6 - Steel, 20 mm x 3 mm x 550 mm
- Strut 7 - Brass, 19 mm x 4.8 mm x 750 mm
- Strut 8 - Aluminium, 19 mm x 4.8 mm x 750 mm
- Strut 9 - Steel, 15 mm x 4 mm x 750 mm
- Strut 10 - Steel, 10 mm x 5 mm x 750 mm

- Strut A - Hardwood, 20 mm x 6 mm x 550 mm with steel knife edge inserts.
- Strut B - Plywood, 20 mm x 6 mm x 550 mm with steel knife edge inserts.
- Strut C - Glass Fibre, 20 mm x 5.5 mm x 550 mm with steel knife edge inserts.
- Strut D - Moulded Brass, D shape, 19 mm x 4.5 mm x 550 mm.
- Strut E - Steel Compound, Two steel struts, bolted together. Both 13 mm x 3 mm. One longer (650 mm) with knife edge ends, the other 640 mm.
- Strut F - Aluminium channel, 13 mm x 13 mm and 1.5 mm thick wall x 750 mm with steel end fittings.
- Strut G - Aluminium angle, 13 mm x 13 mm and 1.5 mm thick wall x 750 mm with steel end fittings.
- Strut H - Aluminium angle, 13 mm x 13 mm and 1.5 mm thick wall x 750 mm with steel end fittings.
- Strut I - Steel Rectangular, 13 mm x 6.4 mm x 650 mm.
- Strut J - Steel Round, 6.4 mm diameter x 650 mm.

Data Recording

Deflection

Time Time (s)	Deflection Indicator		Specimen Properties				DL1 Load Meter							
	Deflection (mm)	Deflection Position (mm)	Material Description	Young's Modulus (GPa)	Second Moment of Area (mm ⁴)	Length (mm)	End Conditions	Force (N)	Peak Force (N)					
Data Series 1														
	0	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	0.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	1	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	1.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	2	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	2.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	3	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	3.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0				
	4	0.00	500	Steel	207	0.00	750	Pinned-Pinned	0	0				
	4.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	5	0				
	5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	9	0				
	5.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	9	0				
	6	0.00	500	Steel	207	0.00	750	Pinned-Pinned	10	0				
	6.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	11	0				
	7	0.07	500	Steel	207	0.00	750	Pinned-Pinned	13	0				
	7.5	0.10	500	Steel	207	0.00	750	Pinned-Pinned	16	0				
	8	0.10	500	Steel	207	0.00	750	Pinned-Pinned	19	0				
	8.5	0.12	500	Steel	207	0.00	750	Pinned-Pinned	20	0				
	9	0.13	500	Steel	207	0.00	750	Pinned-Pinned	21	0				
	9.5	0.13	500	Steel	207	0.00	750	Pinned-Pinned	21	0				
	10	0.18	500	Steel	207	0.00	750	Pinned-Pinned	25	0				
	10.5	0.21	500	Steel	207	0.00	750	Pinned-Pinned	26	0				
	11	0.21	500	Steel	207	0.00	750	Pinned-Pinned	28	0				
	11.5	0.22	500	Steel	207	0.00	750	Pinned-Pinned	28	0				
	12	0.28	500	Steel	207	0.00	750	Pinned-Pinned	30	0				

..... 13 sets of valid data

Strut	1	2	3	4	5	6
Critical Load (N)						

- may not reach buckling load
- load fell for strut 4



Figure 7: Deflection of struts

abnormal

Strut	1	2	5
Critical Load (N)	Actual		
	Other side		

- P_{crit} not the same



Figure 8: Deflection of struts in both side

- could not reach P_{crit}
- stiffness reduced as rate of deflection reduced



Figure 9: Deflection of struts with eccentric

Measurement

- Strut 1 Steel
- Strut 3 Steel
- Strut 5 Steel
- Eccentricity Fitting Steel
- Strut F - Al Channel Aluminium

Data Analysis

$\frac{1}{L^2}$ vs P_{crit}

Strut		1	2	3	4	5	6
Critical Load (N)	Actual						
	Other side						
	Theory						

- E in user guide might inaccurate
- A is different along the strut
- load might too fast

Figure 10: $\frac{1}{L^2}$ vs P_{crit}

Mode of failure

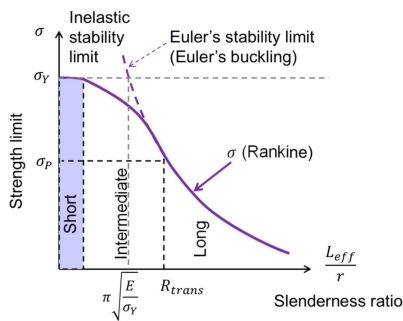


Figure 11: $\frac{L_{eff}}{r}$ vs σ for real struts[2]

Strut	1	2	3	4	5	6
R_{trans}						
Slenderness ratio						
Mode of failure						

- all $\frac{L_{eff}}{r} > R_{trans}$
- strut experience buckling failure

Southwell plot

Deflection, v_c (m)



Figure 12: Southwell plot

- no plots cross the origin e exists without adding it

Buckling Load (N)	Strut 1	Strut 2	Strut 3	Strut 4	Strut 5	Strut 6
Theoretical						
Southwell Plot Method						
Percentage difference (%)						

$P_{crit}^{southwell} < P_{crit}^{theory}$

+ imperfection

+ alignment error

• do not need to reach P_{crit}

• could obtain imperfections

Figure 13: Southwell plot for Strut 1

Figure 14: Southwell plot for Strut 5

Eccentricity e (mm)	Strut 1		Strut 5	
	Small	Big	Small	Big
Measured from Southwell plot				
Actual known offset				
Difference				

$e_{measured} > e_{actual}$

→ existing e in the strut itself